MOBILE DRILL BRAIN

Calibration Technician Training Module 004

Field Setup and SPT Analyzer Software Workflow

Document type	Workspace 006-B technician training module
Module	004 - Field Setup and SPT Analyzer Software Workflow
Version	v0.1 SME-Edited Deliverable
Prepared	July 2026
Source basis	Calibration Data Collection Cheat Sheet; SPT Introduction Draft; Chapter 2 Field Lessons; Module 002 kit readiness; Module 003 SPT fundamentals; SME clarifications captured during review.
Training lane	Calibration Technician Training - software workflow and field setup bridge
Status	Internal training draft; not a final field SOP or analyzer manufacturer manual

Locked module purpose

Teach a calibration technician how the physical field setup connects to the SPT Analyzer software workflow before, during, and immediately after data collection.

This module does not replace hands-on training. It is meant to prevent the common mistake where a technician understands the drilling/calibration process but gets lost in the software sequence.

The calibration cheat sheet is treated as a software workflow reminder for a trained field technician, not as a complete field SOP.

1. Source basis and training boundary

This module is based on current calibration training source material and SME clarification.

- Calibration Data Collection Cheat Sheet - SPT Analyzer button sequence and repeat-depth workflow.
- Module 002 - analyzer kit, cables, instrumented rods, accelerometers, connector care, and job readiness.
- Module 003 - SPT fundamentals, N-value, N60, ASTM context, sampler intervals, and refusal foundation.
- Chapter 2 Field Lessons - sensor tightness, refusal, hammer stroke/RPM, rod connections, and jobsite control.
- SME clarification - CollectWire sequence, pulse check, pause/play behavior, LP/LE definitions, production caveat, and instrumented rod protection.

Training boundary

This module teaches the field-to-software workflow. It does not fully teach advanced data review, final report building, full troubleshooting hierarchy, engineering interpretation, manufacturer theory, customer scheduling, or pricing. Those belong in later modules.

2. Learning outcomes

After completing Module 004, the trainee should be able to:

1. Explain why field setup must be complete before activating data collection.
2. Open the SPT Analyzer software and reach the project/setup screen.

3. Enter rig, customer, borehole, depth interval, LP, and LE information correctly.
4. Define LP as length of penetration and explain why LE must be measured.
5. Use the Sensor tab to confirm sensor connection.
6. Explain why CollectWire is clicked more than once during setup.
7. Explain what the pulse check does and why the graph shape matters.
8. Recognize that the pause button activates live data collection and the play button ends collection.
9. Record blow counts on paper for every 6-inch increment.
10. State the critical warning: never allow the instrumented rod to enter the hollow stem auger or borehole.

3. Big-picture workflow

Track	What it means	Why it matters
Physical setup	Rod string, sampler, instrumented rod, anvil, hammer, sensor cables, and jobsite positioning.	Bad physical setup creates bad data or damaged sensors.
Software setup	Project details, rig ID, depth interval, LP, LE, sensor verification, file naming, and data collection screen.	Bad software setup can make good field data unusable.

The analyzer software does not fix bad setup. The software only records what the field setup sends into it.

The technician controls data quality, but must also work within the project sampling requirements and drilling workflow.

4. Before opening live data collection

Before the sensors are activated, the technician should confirm the setup is physically and administratively ready.

- Correct rig is being calibrated.
- Rig model and serial number are known.
- Customer/company name, borehole ID, driller/operator name, and technician name are known.
- Sampler interval and starting depth are known.
- Depth interval is entered correctly in feet.
- LP is entered correctly.
- LE has been measured.
- Correct rod type is being used.
- Sensor cable is connected and accelerometers are secure.
- Instrumented rod is attached correctly.
- Anvil is threaded flush into the top of the instrumented rod.
- Automatic hammer is lowered onto the anvil only when setup is ready.

Training rule

The pause button must not be pressed until physical setup is complete, the pulse check has passed, and the driller/operator is ready for collection.

5. Opening the SPT Analyzer software

Step	Action	Training note
1	Turn on the SPT Analyzer and allow it to load to the desktop screen.	Do not rush software startup.
2	Click the PDA-S icon on the main screen.	This opens the analyzer software.
3	Wait for the orange SPT screen, then click CollectWire.	This moves the technician into the setup area where rig/project information is entered.

6. Project detail and file setup screen

The project detail screen is where the technician enters the information that names and organizes the data file. The data file is created before collection, and the software automatically routes the data into the designated automatic data folder on the analyzer hardware.

Field	What to enter	Training note
Rig ID	Company, rig model, and rig serial number.	Example format: customer + model + serial.
Depth interval	Starting depth and ending depth of sampler interval.	Entered in feet; must match actual interval.
Borehole ID	Borehole number or identifier.	Example: Bore Hole ID 1.
Operator name	Driller and technician names.	Include both where appropriate.

7. Depth interval rule

The depth interval field tells the software where the sample starts and how far the sampler travels into the soil. It is entered in feet.

Example	Meaning
20.0 to 21.5 ft	18-inch sampler interval.
20.0 to 22.0 ft	24-inch sampler interval.

An 18-inch sampler is the minimum length for proper SPT data collection. A 24-inch sampler is common and often preferred when available because it collects more overall data and makes back-end report math easier.

Technicians will see both 18-inch and 24-inch intervals in the field. The important rule is that the depth interval entered in the software must match the actual interval being collected.

Production drilling caveat

On a production drilling jobsite, calibration should control the workflow as much as reasonably possible, but calibration does not override the sampling standards or interval requirements set by the entity paying for the drilling.

Different projects may require different sample intervals and sampling frequencies. The calibration technician must work within those project requirements.

If the production sampling requirement is met but insufficient calibration data has been collected, the calibration workflow can return to standard data collection afterward to gather enough usable data.

Mixed-interval reporting rule

Different depths and different sample intervals can still be used in one cohesive calibration report as long as depth interval, sample interval, LP, LE, blow counts by 6-inch increment, and field notes are correctly recorded.

8. LP and LE entry

After project details are entered, go to the ROD tab and enter LP and LE.

Term	Meaning	Training rule
LP	Length of penetration. Distance from the top of soil / ground surface to the bottom of the hole.	LP must match the actual hole depth context for the data being collected.
LE	Measured distance used by the analyzer from the bottom-of-hole context to the instrumented rod sensor location.	LE must be measured. Do not guess or copy a prior value blindly.

A common LE value may be around 3.5 ft based on how rod lengths and the instrumented rod setup usually work out, but that value must still be confirmed for the setup being used.

9. Sensor tab connection check

After LP and LE are entered, go to the SENSOR tab. Connect the sensor cable to the instrumented rod system. The sensor boxes/checkmarks should fill in when the cable and sensors are properly connected.

Training rule

Do not proceed to live data collection until the Sensor tab confirms sensor connection.

If checkmarks do not appear, stop and check:

- Cable connection.
- Pigtail connection.
- Accelerometer connection.
- Loose sensor hardware.
- Damaged or dirty connector.

10. Why CollectWire is clicked again

After project information, LP, LE, and sensor connection are confirmed, click CollectWire to confirm setup and move forward.

CollectWire is not a random repeat click. In this workflow, it confirms the setup path and moves the technician toward the actual data collection screen.

CollectWire point	Purpose
After orange SPT screen loads	Moves into the setup area where rig/project information is entered.
After setup and sensor confirmation	Confirms setup and moves the technician toward the data collection screen.
Next CollectWire step	Moves into the actual data collection screen.

11. Pulse check

The button/icon may look like a top hat because it visually represents the shape the graph should show. For training purposes, this step is called the pulse check.

When the pulse check is pressed, the analyzer sends an electrical pulse through the sensors and back to the software.

Result	Meaning	Required action
Expected top-hat graph shape	Sensors are connected and signal is returning to the software.	Proceed only after physical setup is ready.
Graph does not show expected shape	One or more sensors may be loose, disconnected, dirty, damaged, or not securely tightened to the rod.	Stop, check connections/tightness, and repeat the pulse check.

Training rule

Do not activate live collection until the pulse check produces the expected graph shape.

12. Critical physical setup before pressing pause

Before pressing the pause button, the technician must confirm physical setup is ready.

- Instrumented rod is threaded snugly to the rod string.
- Anvil is threaded flush into the top of the instrumented rod.
- Automatic hammer is lowered onto the anvil.
- Sensor cable is connected and clear.
- The rod string is not at risk of pulling the instrumented rod into the hollow stem auger or borehole.
- The driller/operator understands collection is about to begin.

Once pause is pressed, the sensors record everything. That is why setup must be correct before pressing pause.

13. Critical warning - protect the instrumented rod

CRITICAL WARNING

Do not let the instrumented rod go into the hollow stem auger or down into the borehole. This is one of the fastest ways to damage sensors.

The technician should closely observe the first 6 blows to make sure the soil is not so loose that the rod and sampler are advancing too quickly.

As a second barrier, the technician must clearly communicate to the operator before collection begins:

Operator instruction

Stop hammer operation immediately if the instrumented rod is in danger of entering the hollow stem auger or borehole.

This warning matters most when the sampler suddenly advances several inches in one blow.

14. Starting live data collection

When the pulse check passes and physical setup is ready, press the pause button. In this software workflow, the pause button activates live data collection.

Training note

The button may feel backward because it looks like a normal pause icon. In this workflow, pressing pause starts active sensor recording.

After pressing pause:

- Sensors are live.
- Hammer blows will be recorded.
- Unnecessary movement or impacts can be captured.
- The technician should control the pace of collection.

The technician should then give the operator a clear start signal.

15. Blow-count recording during collection

During data collection, the technician must record blow counts on paper for every 6-inch increment.

The software records total blow count, but the paper record captures the individual 6-inch increments. The 6-inch breakdown is essential for report building and for recognizing when refusal is becoming a real field issue.

Interval type	6-inch entries to record
18-inch interval	0-6 in.; 6-12 in.; 12-18 in.
24-inch interval	0-6 in.; 6-12 in.; 12-18 in.; 18-24 in.

The software total alone is not enough for complete report context.

16. Refusal awareness during collection

For training purposes, refusal means the sampler does not advance a full 6 inches in 50 blows.

The technician must not treat refusal data as normal calibration data.

If refusal occurs:

- Document the blow count.
- Document the depth interval.
- Document what happened.
- Do not force questionable data into the report path.
- Follow the proper field decision/escalation path.

17. Ending data collection

During active data collection, the pause button changes into a play button.

11. Press the play button to end collection.
12. The End Test box appears.
13. Hit Yes.

Hit Yes whether the data looked clean or questionable. The data file becomes immediately accessible for review after the test is ended. If there were issues, the saved file can be reviewed and the problem can be documented or rerun.

18. Repeat for each depth interval

For each new depth interval, repeat the setup sequence as needed.

Repeat interval rule

Each interval must have the correct depth interval, sample interval, LP, LE, blow-count breakdown, and saved data file context. Do not copy old values without confirming the new interval.

19. Common mistakes

Mistake	Why it matters	Correct behavior
Treating CollectWire clicks as meaningless	The technician may skip setup or enter collection incorrectly.	Understand what each CollectWire step does.
Guessing LE	Incorrect LE affects the data context.	Measure LE.
Misdefining LP	LP must represent length of penetration from ground surface to bottom of hole.	Confirm current depth context.
Pressing pause too early	Sensors record everything after pause.	Complete physical setup first.
Ignoring bad pulse-check graph	A sensor may be loose or disconnected.	Fix sensor/cable issue first.
Letting instrumented rod enter HSA/borehole	This is one of the fastest ways to damage sensors.	Watch first blows and instruct operator to stop if danger appears.
Only relying on software blow total	Report builder needs 6-inch increments.	Record paper blow counts every 6 inches.
Treating refusal as normal data	Refusal data can distort interpretation.	Document and escalate/follow field decision path.
Forgetting production job requirements	Calibration must not override project sampling requirements.	Work within project standards and collect additional calibration data when allowed.

20. Training scenario

Scenario: A technician has entered the rig information and depth interval. The Sensor tab shows checkmarks, and the technician clicks CollectWire to reach the data collection screen. The pulse check does not produce the expected graph shape. The operator says, "It is probably fine, let us just run it."

Correct trainee response should include:

- Do not start live collection.
- Do not press pause yet.
- Explain that the pulse check verifies sensor signal quality.
- Check cable and sensor connections.
- Check accelerometer tightness.
- Repeat the pulse check.
- Only proceed once the graph shape confirms setup is good.

21. Knowledge check

#	Question	Answer key
1	What is the purpose of the first CollectWire click after the orange SPT screen loads?	It moves the software into the setup area where rig/project information is entered.
2	What information helps name/create the data file?	Rig/customer information, rig model/serial, borehole ID, operator/tech names, depth interval, LP, and LE.
3	What does LP mean?	Length of penetration: distance from the top of soil/ground surface to the bottom of the hole.
4	Should LE be guessed if it is usually around 3.5 ft?	No. LE must be measured.
5	What does the Sensor tab confirm?	That the sensor connection is present and recognized by the software.
6	Why is CollectWire clicked after setup?	To confirm setup and move forward toward the data collection screen.
7	What does the pulse check do?	It sends a test pulse through the sensors and verifies the expected graph shape/sensor response.
8	What should the technician do if the pulse-check graph does not look correct?	Stop, check sensor/cable/accelerometer connections, correct the issue, and repeat the check.
9	What does pressing pause do in this workflow?	It activates live data collection.
10	Why are blow counts recorded on paper every 6 inches?	The software records total blows, but report building and refusal awareness require individual 6-inch increments.
11	What is the refusal rule?	The sampler does not advance a full 6 inches in 50 blows.
12	What is the critical instrumented rod warning?	Do not let the instrumented rod enter the hollow stem auger or borehole.
13	What should the technician tell the operator before collection begins?	Stop hammer operation if the instrumented rod is in danger of entering the hollow stem auger or borehole.
14	What does the play button do during active collection?	It ends data collection.
15	What should the technician do when the End Test box appears?	Hit Yes so the data file becomes available for review.

22. Practical signoff checklist

Signoff item	Pass standard	Trainer initials/date
Opens software correctly	Trainee can turn on analyzer, open PDA-S, and reach CollectWire setup path.	
Enters setup fields	Trainee can enter rig ID, depth interval, borehole ID, operator/tech names, LP, and LE.	
Explains LP and LE	Trainee defines LP correctly and states LE must be measured, not guessed.	
Verifies sensors	Trainee uses Sensor tab/checkmarks and explains what they mean.	
Explains CollectWire sequence	Trainee can explain how CollectWire confirms setup and moves forward.	
Performs pulse check	Trainee knows pulse check must pass before live collection.	
Controls pause/play workflow	Trainee knows pause starts collection and play ends collection.	
Protects instrumented rod	Trainee observes first blows and communicates stop condition to operator.	
Records blow counts	Trainee records every 6-inch increment on paper.	
Handles production caveat	Trainee understands calibration works within project sampling requirements.	
Ends test properly	Trainee hits End Test / Yes and understands data becomes reviewable.	

23. SME-needed / next module inputs

No SME items are currently blocking Module 004 as a training draft. The following items belong in later modules:

- Module 005 - LP, LE, depth intervals, sampler intervals, and file accuracy in more detail.
- Module 006 - bad data, refusal, loose sensors, rapid advancement, damaged equipment, and troubleshooting hierarchy.
- Module 007 - report handoff, file review, paper form reconciliation, and documentation package completeness.

Revision history

Version	Status	Change
v0.1	SME-edited deliverable	Created from in-thread Module 004 draft and SME corrections: LP definition, CollectWire wording, pulse-check terminology, instrumented rod protection warning, production drilling caveat, and mixed-interval reporting rule.